UCLouvain

LMAPR2015 : Physics of nanostructures
Applications of a nanostructure
Rapport

Quantum Dots

Students :
Lucas Bogaert
Lucas Janssens

Teachers :
Prof. Flavio Abreu Araujo
Prof. Jean-Christophe Charlier
Prof. Xavier Gonze

November 2, 2026

Contents

1 Introduction **3**

2 Theoretical study **4**
2.1 Quantum confinement 5
2.2 Opto-electronic properties 6
2.2.1 Core/shell structure of quantum dots 7
2.3 The particle in a box model 9
2.4 A glimpse into photon generation 12

3 Production of colloidal quantum dots **13**
3.1 The LaMer Mechanism 14
3.2 The Hot-Injection Method 15

4 Applications **16**
4.1 Colloidal quantum dot photovoltaics 16
4.1.1 Quantum Dot Heterojunction Solar Cells 17
4.1.2 Intermediate Band Solar Cells 20
4.2 Single photon source 24
4.2.1 Principles of single photon generation from quantum Dots 24
4.2.2 Absorption spectra : from bulk to quantum confinement 25
4.2.3 Optical selection rules and light-matter interaction 26
4.2.4 Multi excitons effects, Auger recombination and photon statistics . 27
4.2.5 A challenge for quantum dots : blinking 28
4.2.6 Discussion on single photon generation, materials and limitations . 29

5 Conclusion **30**

List of Acronyms

CB Conduction band

cQD Colloidal quantum dot

DH Depleted hetero-junction

DOS Density of States

ETL Electron transport layer

FF Fill factor

FTO Fluorine-doped tin oxide

IB Intermediate band

IBSC Intermediate Band Solar Cell

J_{sc} Short circuit current

LDS Luminescent downshifting

MEG Multiple exciton generation

NIR Near-infrared

QD Quantum Dot

QDHSC Quantum dot heterojunction solar cell

QDIBSC Quantum dot intermediate band solar cells

QDSCs Quantum dot-based solar cells

TCO Transparent conductive oxide

UV Ultraviolet

V_{oc} Open circuit voltage

VB Valence band

NIR Near infrared

ITO Indium tin oxide

SPS Single photon source

1 Introduction

For a long time, scientists exploring electronic systems could only study what nature provided, *i.e.* isolated atoms, particles and the ordered structures of metals and semiconductors. Then, in the early 1970s, researchers turned a new page by investigating materials restricted to two dimensions, known as quantum wells. These thin, layered systems revealed properties unlike anything seen before, sparking intense curiosity among physicists and engineers. Laboratories around the world soon joined the effort, driving rapid progress in fabrication techniques and deepening our understanding of low-dimensional physics. By now, the behavior of these two-dimensional structures is well understood. But in the 1980s, advances in ultra-precise lithography opened the door to even smaller systems—quasi-one-dimensional channels called quantum wires, where electrons could move in only a single direction. The next step in this journey was to trap electrons completely, confining them in every spatial direction to create a tiny, nearly zero-dimensional system: the quantum dot. The first quantum dots were successfully produced by scientists at Texas Instruments [1], and soon afterward, other research groups such as AT&T Bell Laboratories [2] and Bell Communication Research made similar achievements [3].

Because electrons in quantum dots are confined in all three dimensions, these structures behave much like atoms do. What makes them so unique is that their size, shape, energy levels, and even the number of trapped electrons can be precisely tuned. Over the past couple of decades, many experiments on quantum dots have focused on their optical characteristics; how they absorb and emit light, including visible and infrared wavelengths. Their narrow and adjustable emission spectra make them ideal candidates for innovative optical technologies from quantum devices to advanced imaging and display systems. Their emission tunability can be seen in Figure 1, showing that different sized-quantum dots have different emission wavelength.

Colloidal quantum dots have emerged as particularly attractive platforms due to their solution-based synthesis, enabling cost effective production and versatile integration with diverse substrates. Colloidal quantum dots offer room-temperature operation [4], narrow emission linewidth and tunable optical properties. Those are the reasons why cQDs offer practical and technological applications in quantum photonic.
Note : Artificial intelligence tools like ChatGPT and Perplexity were used to rewrite, rephrase and paraphrase a portion of this report.

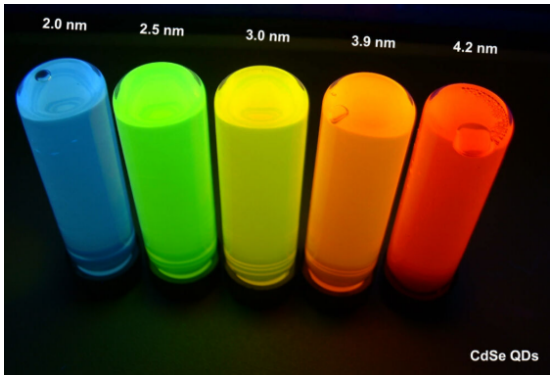


Figure 1: CdSe Quantum Dots. The wavelength of their fluorescence is dependent on the size of the quantum dot [5].

2 Theoretical study

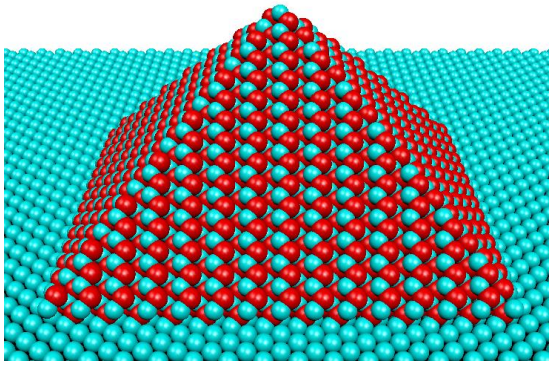


Figure 2: Quantum box made of InAs and GaAs [6].

A quantum dot, also called a quantum box, is a nanometer-sized semiconductor crystal capable of completely confining particles like free electrons. One specific type of quantum dot is shown in Fig. 2. As a matter of fact, due to its small size, the rules of quantum mechanics apply to a quantum dot which behaves like a potential wall confining the electrons inside them. Generally, nanocrystal QDs comprise a few hundred or thousands of atoms and have a size on the order of 1–10 nm. The variation in the diameter leads to changes in the density of states of the semiconductor crystal. Theses changes will lead to various effects discussed further in this report.

As a reminder, when a semiconducting material is illuminated by a photon, an electron in the valence band gets excited to the higher energy band namely the conduction band and leaves behind a vacancy in the valence band called a hole. The electron-hole pair created is bound by Coulomb interactions in what is called an exciton and their spatial separation is defined by the exciton Bohr radius. The exciton Bohr radius can be expressed as:

$$a_B^* = \varepsilon_r \left(\frac{m_e}{\mu} \right) a_B \tag{1}$$

where $a_B = 0.053$ nm is the Bohr radius, m_e is the free electron mass, $\mu = (m_e m_h)/(m_e + m_h)$ is the reduced mass with m_h the mass of the hole and ε_r is the relative permittivity [7]. Then the excited electron can recombine with the hole in the valence band releasing energy in the form of light during a process called photoluminescence. Importantly, the energy of the emitted photon depends on the energy gap between the two discrete energy levels of the valence and conduction band in the QD. Further in this report, it will become clear why we can define discrete energy levels for semiconducting QDs due to their ability to confine charge carriers. Two distinct features of QDs can be put forward:

- The quantum confinement in QDs is similar to the 3D particle in a box model.
- The QDs absorption/emission property corresponds to transitions between discrete quantum mechanically allowed energy levels in the crystal.

Those two properties are what makes QDs look a lot like atoms hence why they are often called artificial atoms. In the two following sections, we investigate further those properties.

2.1 Quantum confinement

In a bulk semiconductor, electrons and holes can move freely through the crystal lattice and their allowed energy levels form quasi-continuous bands (valence and conduction bands) [8]. However, when the diameter of the semiconductor nano-crystal is of the same order as the de Broglie wavelength of the electron wave function, the motion of the charge carriers becomes spatially confined and the energy levels become discrete quantized levels just like in the box model (see Figure 3). Therefore, the exciton (electron-hole pair) can be modeled by the quantum particle in a box model. Consequently, the electron and the hole can be seen as an hydrogen atom with the hydrogen nucleus replaced by the hole of positive charge. Then the energy levels of the exciton can be represented as the solution to the particle in a box at the ground level ($n = 1$) in which the energy of states depend on the length of the box (as we will see in equation 11). Thus by varying the size of the QD, the confinement energy of the exciton can be controlled. Specifically, the confinement effect describes the phenomenon resulting from electrons and holes being squeezed into a dimension that approaches a critical quantum measurement *i.e.* the exciton Bohr radius.

From then on, we can define three regimes [7], depending on the QD's size with respect to the exciton Bohr radius, which specify the wave function of the electrons and holes. In the weak confinement regime, the quantum dot is larger than the exciton Bohr radius, thus the confinement energy is smaller than Coulomb interactions. But the fascinating part about QDs begin in the regime where the exciton Bohr radius and the QD's size (length of the box), linked to confinement potential, are comparable, namely the intermediate confinement regime. In this regime, QDs have intermediate properties between bulk semiconductors and discrete atoms. As the size of the QD approaches the exciton Bohr radius, the excitons get more and more confined and the coulombic energy between electrons and holes becomes weaker than ΔE_{conf} (see equation 4), resulting in free electron and hole states with different bandgaps. This is called the strong confinement regime.

Given that the exciton is treated like an hydrogenoid atom, one can define a reduced effective mass :

$$\mu = \left(\frac{1}{m_e^*} + \frac{1}{m_h^*} \right)^{-1} \tag{2}$$

and the exciton binding energy is given by :

$$E_{\text{exciton}} = -\frac{1}{\varepsilon_r^2} \frac{\mu}{m_e} R_H = -R_H^* \tag{3}$$

with the Rydberg constant $R_H \approx \frac{m_e}{2\hbar^2}(\frac{e^2}{4\pi\epsilon_0}) \approx 13.605$ eV for hydrogen. The energy that is used to confine the charge carriers inside the QD is obtained by using the particle in a box model and is given by :

$$\Delta E_{\text{conf}} = \frac{\hbar^2 \pi^2}{2L^2} \left(\frac{1}{m_e^*} + \frac{1}{m_h^*} \right) = \frac{\hbar^2 \pi^2}{2\mu L^2} \tag{4}$$

In a nutshell, the sum of these energies can be represented by the Brus equation [8] :

$$E_{\text{g,dot}} = E_{\text{g,bulk}} + \Delta E_{\text{conf}} + E_{\text{exciton}} = E_{\text{g,bulk}} + \frac{\hbar^2 \pi^2}{2\mu L^2} - R_H^* \tag{5}$$

Those equations imply that electronic transitions in QDs are inversely proportional to the square of the length of the box i.e the diameter of the QD. Therefore, manipulations

of the size of QDs modify the band gap $E_{g,dot}$ enabling the absorption and emission of specific wavelengths of light. The smaller the QD, the larger the band gap and thus the shorter the wavelength of light that can be absorbed.

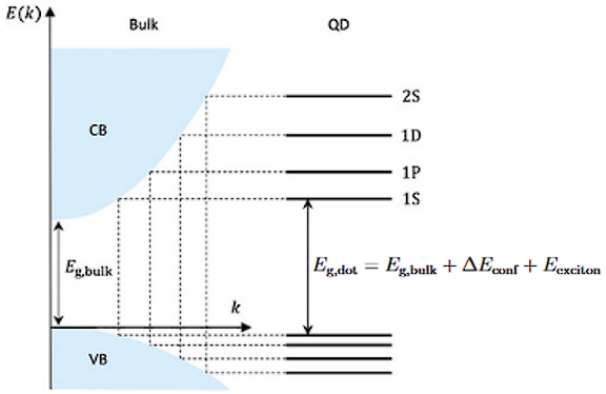


Figure 3: Comparison of the electronic energy structure of a bulk semiconductor and a QD [7].

2.2 Opto-electronic properties

This section will go in further detail on why QDs possess opto-electronic properties that differ from those of larger particles. In addition to the intrinsic band gap energy, inherent to semiconductors $E_{g,bulk}$, the specific emission energy of a QD also depends on its dimensions which leads to the interesting optical property of band gap tunability. Indeed, if we shrink a QD dimension down to a size comparable to the exciton Bohr radius, the exciton is physically confined within the semiconductor resulting in discretizations of energy levels and an increase of the band gap of the semiconducting QD (see Figure 4).

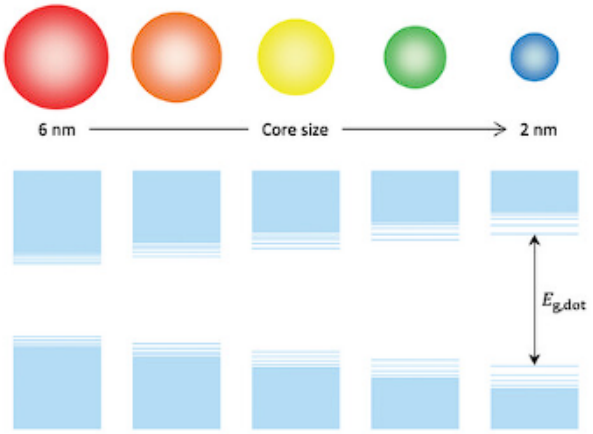


Figure 4: Illustration of band gap tunability in QDs [7].

As the confinement energy depends on the size of QD, both absorption processes and fluorescence emissions can be tuned by changing the size of the quantum dot during its synthesis. The bigger QDs (e.g 5–6 nm diameter) emit longer wavelengths and the smaller QDs (e.g 2–3 nm diameter) emit shorter wavelengths. Moreover, the lifetime of fluorescence emission is also determined by the size of the quantum dot. Indeed,

bigger QDs have more closely spaced energy levels in which the electron–hole pair can be trapped meaning that electron–hole pairs live longer, *i.e.* recombine more slowly in big QDs causing them to show a longer radiative lifetime.

However, the electrons in QDs can undergo a relaxation process to the ground state without any luminescence via non-radiative recombination such as Shockley–Read–Hall recombination, field-induced quenching and Auger recombination. The later will be discussed in Section 4.2.4. Considering that radiative recombination mechanism compete with these other processes, we can determine the ratio of emitted photons to absorbed photons called the quantum yield of QDs based on the recombination rates as described in equation 6 [7].

$$\eta = \frac{(1-N)k_r}{k_r + k_{nr}} = \frac{1-N}{1+\tau_r/\tau_{nr}} \tag{6}$$

with

- N the proportion of QDs in non-emitting states
- k_r the radiative recombination rate
- k_{nr} the non-radiative recombination rate
- τ_r the radiative recombination lifetime
- τ_{nr} the non-radiative recombination lifetime

2.2.1 Core/shell structure of quantum dots

In early studies, QDs were fabricated using a core without a shell [9]. However, in many colloidal QDs, the surface atoms have dangling bonds or are undercoordinated. When exposed to oxygen, the QD surface tends to oxidize, forming a native oxide layer which introduces defects leading to localized energy levels called trap states lying within the bandgap of the semiconductor. These act as so-called recombination centers that compete with radiative recombination. Indeed, these states facilitate carrier trapping and recombination through phonon emission rather than photon emission, thereby reducing the quantum yield. Therefore, the sole core structure suffers from serious surface recombination, thus necessitating a protective layer to enhance stability and minimize surface defects. In order to improve the quantum yield and obtain a high performance, QDs can be made of a core/shell structure to confine the electrons and holes within the core and thus have a more efficient radiative recombination rate [10]. This improvement is induced by the reduced access of electron and hole to non-radiative surface recombination pathways but also by the reduced Auger recombination. In addition to the enhanced charge confinement effect and reduced Auger recombination, the gradient shell composition (band edges change smoothly with radius) effectively decreases interfacial lattice mismatch-induced stress and tailors the energy level of QDs [10]. Indeed, if the core and shell lattice parameters do not match well, additional epitaxial shell growth becomes laborious resulting in the formation of interfacial defects and trap states.

The core/shell structure can be categorized as type I, type II or quasi-type II distinguished by the alignment of their conduction and valence bands (see Figure 5). In the type I structure, the bandgap of the core is smaller than the one of the shell and the respective conduction band minimum and valence band maximum of the core are within

the bandgap region of the shell so that both electrons and holes are effectively confined. Type II structures possess an energy level in which both band edges of the shells are lower or higher than those of the core, creating an asymmetric wave function [7][10]. Although the shells in type II QDs do not confine the charges as well as in type I, they provide the advantage of an exciton lifetime that is longer than the one of type I QDs because the electron and hole are spatially separated across the core–shell interface, which reduces their wavefunction overlap. Indeed, since radiative recombination depends on this overlap, a reduced overlap means slower recombination and therefore a longer exciton lifetime. This long exciton lifetime makes the type II structure an attractive structure for photovoltaic applications [7]. Quasi-type-II QDs differ from type II QDs in that they have only partial separation, typically because one carrier is tightly confined in the core while the other is delocalized and its wavefunction spreads into the shell.

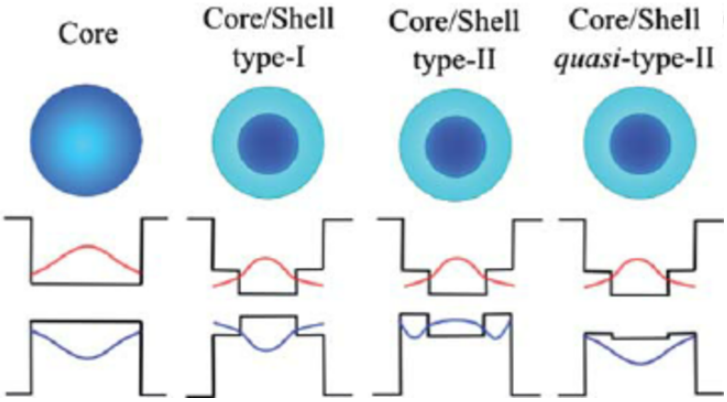


Figure 5: Band gap structure for different types of QD core/shell structure [10].

Property	**Type-I**	**Quasi-Type-II**	**Type-II**
Electron–hole separation	None	Partial	Strong
Carrier locations	Both in core	One in core, the other delocalized	Electron & hole in different regions
Recombination rate	Fast	Medium	Slow
Exciton lifetime	Short	Medium	Long
Emission brightness	High	High–medium	Low
Applications	LEDs, displays	Lasers, SPS	Photovoltaics

Table 1: Table of properties for different types of QDs.

The properties of QDs observed in this section, namely the size-dependent band gap tunability, the quantum yield control and the behavior of different QD core/shell structures all stem from the spatial confinement of charge carriers. To quantitatively predict and understand how this confinement determines the energy spectrum of a quantum dot, we must develop an explicit mathematical model. The particle in a box model provides exactly this framework. This model describes the quantum confinement effect by solving the Schrödinger equation for a particle trapped within a potential well with boundary conditions that reflect the nanoscale dimensions of QDs. Since electrons in quantum dots are confined in all three spatial dimensions simultaneously, we must solve this model in three dimensions. By doing so, we will derive the discrete energy levels that result from

confinement, demonstrate how these levels depend on the size of QD, and explain why the density of states transforms from the continuous function observed in bulk semiconductors into discrete delta functions. Ultimately, the particle in a box model rationalizes the empirical Brus equation introduced earlier and provides physical insight into the optical phenomena that makes quantum dots technologically relevant.

2.3 The particle in a box model

The unique optical properties of quantum dots, *i.e.* their size-tunable band gap, the emergence of discrete optical transitions, and modified lifetimes all stem directly from quantum confinement that we will now describe. To explain why confining electrons to nanoscale dimensions produces discrete energy levels rather than continuous bands, we turn to the particle in a box model, a foundational approach in quantum mechanics. In QDs, this model operates in all three spatial dimensions simultaneously. By solving the Schrödinger equation within a finite potential well and comparing predictions to experiments, we will show how this model accounts for the changes in opto-electronic properties observed in the previous section.

In QDs, we have two components affecting how the quantum wells form physically. Firstly, in a colloidal QD or epitaxially grown QD, there is the bandgap difference between the QD material and its surrounding shell that causes the conduction band discontinuity characterized by quantum wells. This property creates a potential well for both electrons and holes (see Figure 5). Hence why we use the effective mass of the charge carriers to describe the quantum well because the value of effective mass generally differs between different materials as it depends on the curvature of the band. Secondly and most importantly, the small size of the QD acts as a potential well due to the quantum confinement effect. The model of the particle in a box is used to make predictions about what sort of behavior quantum wells are going to exhibit. This model describes the situation where a particle (e.g an electron) is stuck in a potential well and so it cannot escape because the potential barrier V_0 is extremely high (see Figure 6). The walls of this one dimensional box may be seen as regions of space with an infinitely large potential energy [11]. Assuming that charge carriers experience no potential within the box, the time independent Schrödinger equation reduces to, with m_e^* the effective electron mass :

$$-\frac{\hbar^2}{2m_e^*}\frac{d^2\psi(x)}{dx^2} = E\psi(x) \tag{7}$$

which means the wave functions inside the box are the solution of a harmonic oscillator :

$$\psi(x) = A\sin(k_n x) + B\cos(k_n x) \tag{8}$$

We know that wave functions cannot exist in the barrier region of the box due to the infinitely high potential hypothesis. Therefore, we can impose the following boundary conditions :

$$\psi(0) = \psi(L) = 0 \tag{9}$$

And the solution to the Schrödinger equation is a wave function taking the following form

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin(k_n x), \quad k_n = \frac{n\pi}{L} \tag{10}$$

where the prefactor was found by normalizing the wave function. The subscript n ($n > 0$) is the integer quantum number and k_n is the wave number associated with each quantum state. What is interesting in this model is that it makes predictions about the discrete energy levels that the electrons are allowed to have. Indeed, the energy associated to the wavefunction is given by :

$$E_n = \frac{\hbar^2 k_n^2}{2m_e^*} = \frac{\hbar^2}{2m_e^*}\left(\frac{n\pi}{L}\right)^2 = \frac{\hbar^2\pi^2}{2m_e^* L^2}n^2 \quad n = 1, 2, 3, \ldots \tag{11}$$

By comparing this result to our earlier observation in Section 2.2 that smaller QDs emit shorter wavelengths, the energy scaling $\propto 1/L^2$ directly explains why halving the dot diameter quadruples the transition energy and thus blueshifts the emission. We also notice

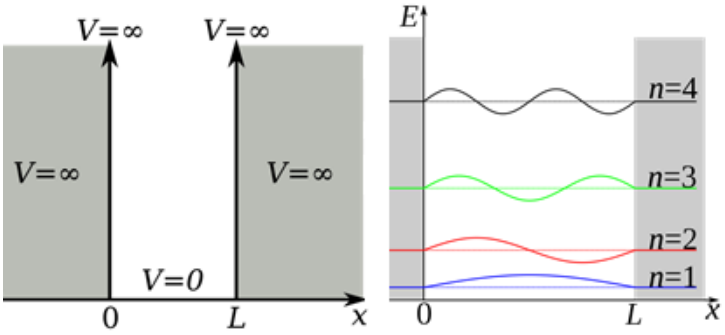


Figure 6: Particle in a box model with infinite barriers and wavefunctions corresponding to each energy [12].

that the energy levels increase with n^2 meaning that high energy levels are separated from each other by a greater amount than low energy levels are. We also see that the ground state always has a positive energy contrarily to a classical system.

This is a good model in the case of QDs because electrons need a relatively high energy (a couple eV) in order to be able to escape from the semiconductor but no electrons have that amount of energy at room temperature. It is also important to notice that because electrons are confined in 3D in QDs, we have quantum wells in all three directions of space hence their behavior can be described by the three dimensional particle in a box energy quantization equations. For a three dimensional box, the approach is similar to the 1D case and we have the solutions:

$$\psi_{n_x,n_y,n_z} = \psi_{n_x}(x,t,L_x)\,\psi_{n_y}(y,t,L_y)\,\psi_{n_z}(z,t,L_z), \tag{12}$$

$$E_{n_x,n_y,n_z} = \frac{\hbar^2 \mathbf{k}^2_{n_x,n_y,n_z}}{2m_e^*}, \tag{13}$$

where the three-dimensional wavevector is given by:

$$\mathbf{k}_{n_x,n_y,n_z} = k_{n_x}\,\hat{\mathbf{x}} + k_{n_y}\,\hat{\mathbf{y}} + k_{n_z}\,\hat{\mathbf{z}} \tag{14}$$

$$= \frac{n_x\pi}{L_x}\,\hat{\mathbf{x}} + \frac{n_y\pi}{L_y}\,\hat{\mathbf{y}} + \frac{n_z\pi}{L_z}\,\hat{\mathbf{z}}. \tag{15}$$

One drawback of this infinite well model is that it predicts many more energy states than exist as the walls of real quantum wells are finite. The model is not as accurate as

we would want because it also neglects the fact that in reality the wave functions do not go to zero at the boundary of the box. Indeed, there is a non zero probability of finding charge carriers out of the box due to quantum tunneling. To be more accurate, we can instead use the finite potential model [11] where the walls of the box are modeled using a finite potential V_0. In this case, since the walls are finite, the electrons can tunnel into the barrier region. Therefore, the allowed wave functions will penetrate the barrier wall but will decay exponentially to zero as we go away from the well (see Figure 7).

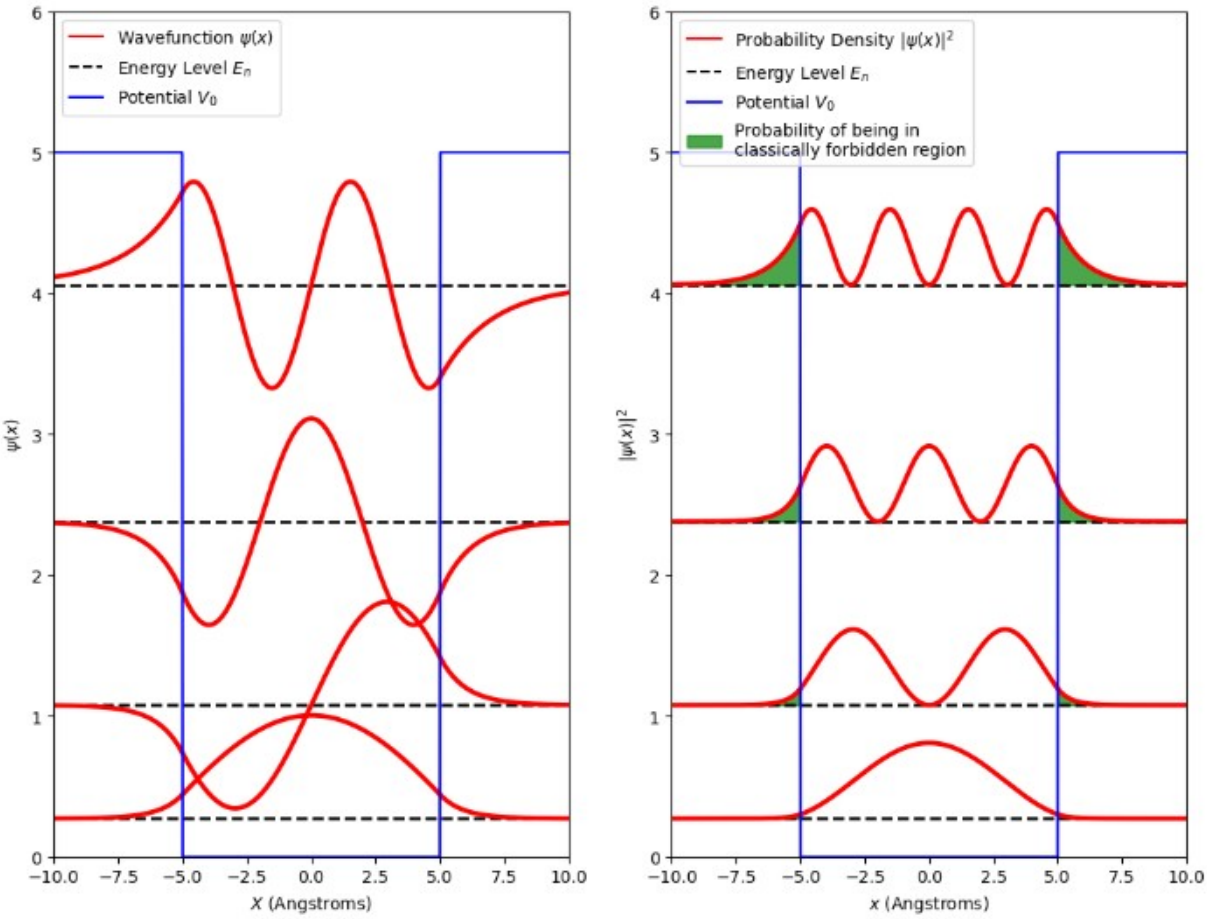


Figure 7: Particle in a box model with finite barriers. Illustration of the wavefunctions (left) and probabilities (right) associated to specific energy levels [11].

So far, we have seen that, in a quantum dot; the electronic, electrical and optical behavior of the carriers are governed by quantum mechanics. Also, energy quantization, due to electron confinement in a region of space, can have a great impact on materials in terms of the electronic density of states (DoS). Let's now investigate how these quantum effects can affect the DoS. The carrier density in the material for different dimensions of confinement is presented in Table 1 and can be seen in the Figure 8.
The δ-function density of states in QDs (contrasting with the continuous spectrum in bulk materials) directly explains the absorption peaks observed experimentally in Figures 24 and 25 of Section 5.2.2, and justifies why QDs exhibit narrow, discrete emission lines unlike their bulk counterparts.
As seen before, discrete states are produced by quantum confinement and obey the previously mentioned Schrödinger equation. Since all three dimensions are confined, the

Structure	Limited Number of Dimensions	dN/dE
Bulk material	0D	$\sqrt{E}$
Quantum well	1D	step
Quantum wire	2D	$1/\sqrt{E}$
Quantum dots	3D	$\delta(E)$

Table 2: Density of states dN/dE as a function of dimensional confinement.

density of states is a series of delta functions. The DoS is given by the expression

$$\rho_{0D} = \delta(E - E_{n,m,q}) \tag{16}$$

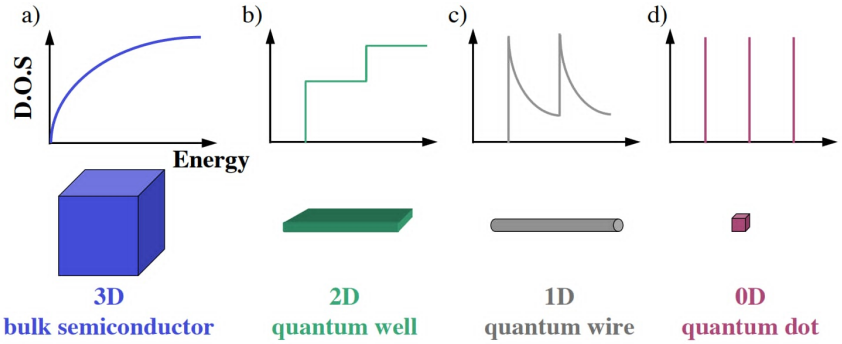


Figure 8: DoS for different dimensional confinement [13].

2.4 A glimpse into photon generation

The process of photon generation by QDs will be discussed in depth in Section 5.2. In short, the mechanism behind which quantum dots are able to emit light relies on the formation of so-called quasiparticles, called excitons. We now turn to a quantum-mechanical calculation of the exciton binding energy [14]. The result obtained in this section will help us understand how the absorption spectra of QDs are related to their size.

For spherical QDs (of radius R), like colloidal quantum dots and without considering Coulomb interaction, the Schrödinger equation of an electron or hole can be written as

$$-\frac{\hbar^2}{2m_i}\nabla^2\Psi_i(r) = E_i\Psi_i(r) \tag{17}$$

Where i is electron or hole. Under ideal quantum constraints, the boundary condition is

$$\Psi_i(r) = 0 \quad \text{for} \quad r = R \tag{18}$$

With this boundary condition, the Schrödinger equation is solved with

$$\Psi_i(r) = \sqrt{\frac{1}{4\pi R^3}}\frac{j_l(\alpha_{nl}r/R)}{j_{l+1}(\alpha_{nl})}Y_l^m(\theta,\phi) \tag{19}$$

Here j_l is the l-order spherical Bessel function and α_{nl} is the n-th root of the l-order spherical Bessel function. Using the Bessel function, boundary condition can be written as

$$j_l(\alpha_{nl} r/R)|_{r=R} = 0 \tag{20}$$

That is $j_l(\alpha_{nl}) = 0$
By substituting the form of the wavefunction into Schrödinger equation, it is possible to obtain discrete eigenvalues,

$$E_i = \frac{\hbar^2}{2m_i}\left(\frac{\alpha_{nl}}{R}\right)^2 \tag{21}$$

Hence, the energy levels of electrons and holes are

$$E_e = E_g + \frac{\hbar^2}{2m_e}\left(\frac{\alpha_{n_e l_e}}{R}\right)^2 \tag{22}$$

$$E_h = -\frac{\hbar^2}{2m_h}\left(\frac{\alpha_{n_h l_h}}{R}\right)^2 \tag{23}$$

For the electron, the energy is measured relative to the conduction band edge E_g, while the energy of the hole is negative relative to the valence band edge set at zero, reflecting its position within the valence band.

But so far, this development dit not take into account the Coulomb interaction between electrons and holes. The corrected-Schrödinger equation that describes the electron-hole pair is

$$\left(-\frac{\hbar^2}{2m_e}\nabla_e^2 - \frac{\hbar^2}{2m_h}\nabla_h^2 + V_c\right)\Phi(r) = E\Phi(r) \tag{24}$$

This equation is given in the spherical coordinates with the boundary condition $\Phi(r = R) = 0$ and V_c is the Coulomb potential. The Schrodinger equation (24), along with the corresponding boundary condition, can be solved analytically in the absence of the Coulomb interaction, resulting in the electron-hole pair energy

$$E = E_g + \frac{\hbar^2}{2m_e}\left(\frac{\alpha_{n_e l_e}}{R}\right)^2 + \frac{\hbar^2}{2m_h}\left(\frac{\alpha_{n_h l_h}}{R}\right)^2 \tag{25}$$

This equation shows that the absorption is blue-shifted with respect to the bulk bandgap E_g. The shift varies like $1/R^2$, with crystal size R, being larger for smaller sizes.

The theoretical framework established in this section reveals that quantum dot properties depend critically on their size through quantum confinement effects [13] [15] [16]. However, creating monodisperse quantum dots with precisely controlled dimensions requires an understanding of the chemical and physical mechanisms governing their synthesis. The following section explores how colloidal synthesis techniques, driven by the LaMer nucleation model and implemented through methods such as hot-injection, enable controlled production of quantum dots whose dimensions can be tuned to match the energy level predictions of the particle in a box model.

3 Production of colloidal quantum dots

Colloidal quantum dots are a specific class of quantum dots that are chemically synthesized and dispersed in a liquid medium before being applied or deposited for practical

use. During colloidal synthesis, reactive precursors are mixed in solution, leading to the nucleation and growth of solid nanocrystals suspended in the liquid. This process continues until the nanocrystals reach the desired size, at which point the reaction is stopped or stabilized. Once purified, the quantum dots are re-dispersed in an appropriate solvent for subsequent processing or application.

3.1 The LaMer Mechanism

The foundational framework for understanding colloidal quantum dot synthesis originates from the LaMer nucleation model [17]. It is named after Victor K. La Mer who first proposed the model in 1950 [18]. The process involves three distinct stages; first, the concentration of monomers rapidly increases to a supersaturated level (Stage 1) through precursor decomposition or injection, leading to a "burst" of nucleation (Stage 2), which quickly consumes the monomers. Finally, the monomer concentration drops below the supersaturation level, and the existing nuclei grow by consuming monomers via diffusion-controlled growth (Stage 3), as you can see on Figures 9 and 10. Figure 11 is an actual TEM image of a chemically produced QD.

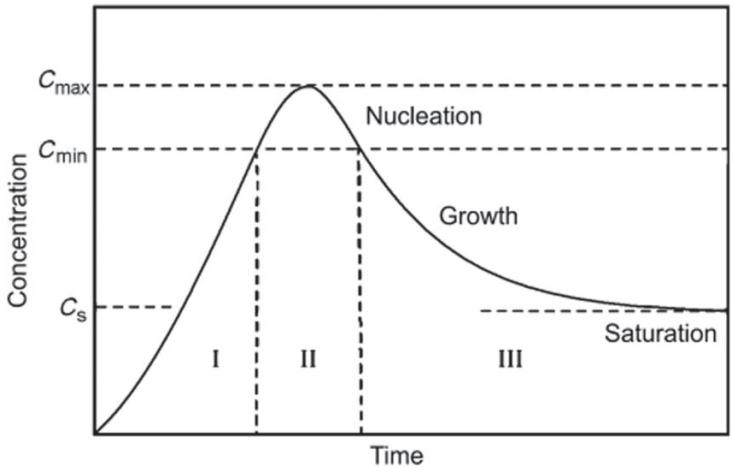


Figure 9: LaMer mechanism for nanoparticle growth [19].

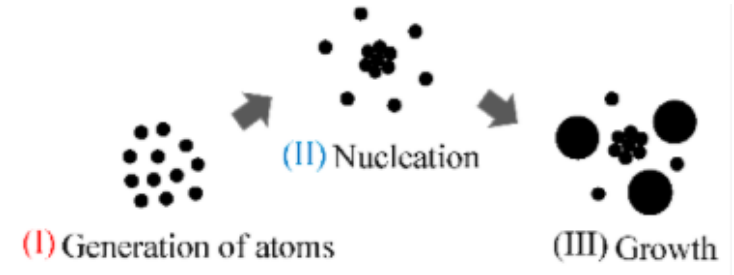


Figure 10: The three steps of LaMer mechanism [20].

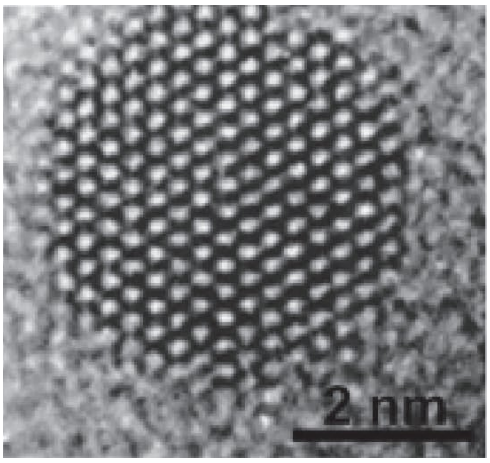


Figure 11: TEM image of an array of atoms forming a single CdSe QD produced using the LaMer mechanism [21].

3.2 The Hot-Injection Method

The hot-injection method [17], pioneered by Bawendi and Alivisatos in the 1990s, remains the most widely used approach for synthesizing monodisperse quantum dots (see Figure 12). The physical principle underlying this technique is the generation of homogeneous supersaturation through rapid injection of room-temperature organometallic precursors into a high-temperature ($\approx$ 300K) solvent containing coordination ligands. Supersaturation occurs when a solution contains more dissolved solute than its equilibrium solubility allows. This metastable state provides the chemical potential needed for new nuclei to form and grow into nanoparticles. The injection generates a sharp increase in monomer concentration, which initiates an intense but short-lived, *i.e.* burst nucleation episode. Nucleation is a critical step where individual particles (atoms, ions,...) aggregate to form a stable cluster (=nucleus), which then acts as a seed for further growth. The rapid temperature drop following injection shifts the system from a nucleation-dominated regime to a growth-dominated regime. This thermal quench is actually extremely important; it depletes the monomer concentration below the critical nucleation threshold while maintaining sufficient conditions for continued growth of existing nuclei [21][22].

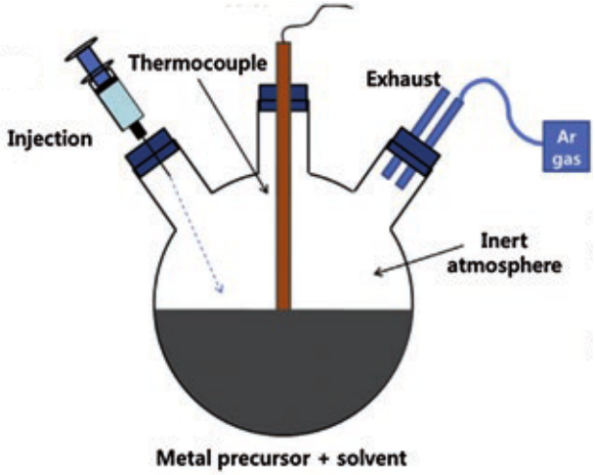


Figure 12: The experimental setup of the hot injection method [23].

4 Applications

4.1 Colloidal quantum dot photovoltaics

The growing global demand for sustainable and renewable energy sources has intensified research into next-generation photovoltaic technologies that can overcome the efficiency limitation of traditional silicon-based solar cells. Indeed, crystalline Si is a relatively weak light-absorber due to its indirect bandgap and so it requires thick high-purity solar cells. Though, the bandgap of crystalline silicon (1.12 eV) is very close to the optimum theoretical bandgap (1.34 eV) predicted by the Shockley–Queisser limit for single-junction solar cells. As a result, QDSCs aim to overcome this efficiency limitation by exploiting quantum effects such as MEG and tunable bandgaps. Indeed, cQDs are extremely promising for next-generation photovoltaic technologies due to their unique optical properties such as size-tunable emission wavelength, antireflective features, luminescent downshifting and multiple exciton generation.

The property of size tunability of the bandgap in QDs enables the design of multi-junction or spectrum-splitting solar cells in which different layers or sets of QDs are optimized to absorb distinct portions of the solar spectrum. By matching each QD layer to a specific energy range of sunlight, these architectures are able to maximize photon utilization and significantly reduce thermalization losses leading to higher overall photovoltaic conversion efficiencies compared to conventional single-junction cells.

cQDs also provide anti-reflective benefits at longer wavelengths. As a result, they reduce optical losses due to reflection and enhance the absorption of NIR light. The antireflective effect arises mainly from two mechanisms [24]. Firstly, cQDs deposited as thin films often form porous layers which leads to an effective refractive index gradually changing between air and the underlying substrate. Such a refractive index gradient reduces Fresnel reflection at the interface, particularly at longer wavelengths which corresponds to electromagnetic waves with high optical penetration. Secondly, aggregates with sizes comparable to the wavelength of light within the QD layer can introduce scattering. This scattering of light increases the effective optical path length, thereby enhancing light trapping and reducing reflection losses.

Although high-energy photons can be easily absorbed in conventional solar cells, the resulting electron–hole pairs are often generated very close to the semiconductor surface, where they are susceptible to strong recombination losses. This issue can be mitigated by using cQDs to transform high-energy photons into lower-energy photons that can be used effectively by the solar cells thereby reducing surface recombination losses and enhancing the absorption of UV light; this process is called luminescent downshifting [25]. LDS arises naturally from the discrete energy levels in QDs and their ability to absorb high-energy photons and re-emit lower energy photons without losing much energy through phonons emissions.

The antireflective effect and LDS contribute to the broad absorption spectrum of QDs (spanning from the UV to the NIR regions) by enabling the capture of a wider portion of the solar spectrum compared to conventional semiconductors. As a result, this large spectral coverage allows for more efficient photon harvesting which leads to an increase in the generation of electron–hole pairs over a wide range of wavelengths. Consequently, the overall solar energy conversion efficiency in QD photovoltaic devices is improved.

The phenomenon of MEG allows a single photon absorbed by a QD to produce two or more electron–hole pairs instead of just one as in conventional semiconductors used

in solar cells. As a result, MEG increases the number of charge carriers generated per absorbed photon thereby enhancing the photocurrent and raising the theoretical maximum conversion of sunlight into electricity of the solar cell beyond the Shockley–Queisser limit. This theoretical limit constrains single-junction photovoltaic devices with an ideal band gap of 1.34 eV to a maximum efficiency of about 33.7% where the ideal band gap is the energy level that maximizes the efficiency for a given solar spectrum. The Shockley–Queisser limit accounts for the primary ways a solar cell loses energy. Firstly, there is spectral losses where photons with energy below or above the band gap cannot be absorbed. Then the radiative recombination which corresponds to the process where an electron and a hole recombine to emit a photon. Finally, the process where the excess energy of high-energy photons is lost as heat called thermalization is also taken in account. While real-world solar cells have additional losses due to non-radiative recombination, the Shockley-Queisser limit provides a benchmark for evaluating solar cell performance.

In this report, we will focus on two types of QD solar cell that helps increasing the overall power conversion efficiency : QD heterojunction solar cells (QDHSC) and QD intermediate band solar cells (QDIBSC).

4.1.1 Quantum Dot Heterojunction Solar Cells

These devices employ cQDs as light absorbers in a heterojunction architecture that facilitates efficient charge separation and collection. QDHSCs typically feature a bulk film of QDs as the main photoactive absorber layer. The advantage of using dense QD films is that they act as strong optical absorbers due to their large absorption coefficients ($>10^5$ cm^{-1}) [25]. Their nanostructured morphology also promotes light trapping and antireflective behavior.

A typical QDHSC consists of the following components:

- Transparent conductive oxide electrode usually made of indium tin oxide or fluorine-doped tin oxide.
- Electron transport layer made of a wide-bandgap n-type semiconductor such as TiO_2, ZnO or SnO_2 is used to facilitate electron extraction from the QDs while blocking holes.
- QD absorber layer made of a thin film (typically 100–300 nm) of closely packed QDs acts as the photoactive layer where light absorption, exciton generation and charge separation occur.
- Cathode made of metals such as Au, Ag or Al serve as the back contact for completing the circuit.

This kind of cQD photovoltaic device can be fabricated on TCOs which serve as optically transparent electrodes. Due to their deep valence band energies and suitable work functions, TCOs act as effective electron-selective contacts. Therefore, it facilitates electron extraction while suppressing undesired hole collection [25]. QDHSCs also rely on the formation of a depletion region at the interface between the QD film and the ETL. This depletion region, established through a p–n heterojunction or a Schottky junction depending on the doping of the QDs and the ETL, provides the built-in electric field necessary for field-driven charge transport and separation.

Moreover, this device employs infrared-bandgap size-effect-tuned PbS cQDs enabling broadband harvesting of the solar spectrum. QDs absorb photons with energies above their bandgap producing electron–hole pairs as follows :

$$\mathrm{QD} + h\nu \rightarrow h^+ + e^- \tag{26}$$

At the QD/ETL interface, energy band alignment facilitates electron injection into the ETL and block undesired hole transfer. The heterojunction thus promotes efficient charge separation and suppresses recombination. Then, the injected electrons spread through the ETL toward the TCO electrode while holes migrate toward the cathode. Finally, the collected electrons and holes flow through the external circuit and by doing so generate a photocurrent and electrical power.

Selecting an appropriate solar cell architecture is essential for maximizing the absorption of the sun's spectrum thus producing a high current density while maximizing the amount of work extracted from each absorbed photon via a large open-circuit voltage and fill factor [25]. This can be reduced to maximizing a single quantity named the power conversion efficiency η which is given for an input solar intensity P_{solar} (typically 100mW cm^{-2}) by :

$$\eta = \frac{V_{oc} J_{sc} FF}{P_{solar}} \tag{27}$$

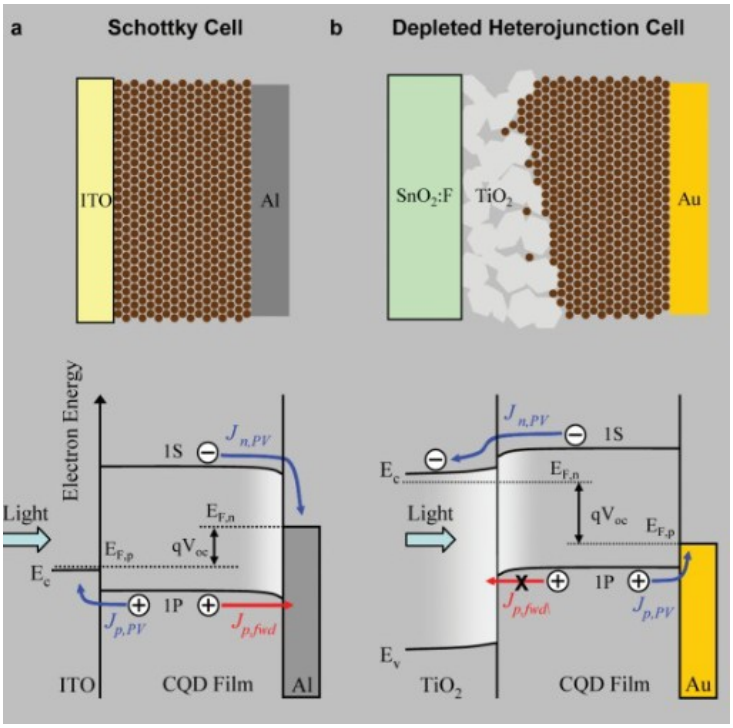


Figure 13: Two cQD photovoltaic architectures under photovoltaic operation close to maximum V_{oc}. $E_{F,n}$ and $E_{F,p}$ are the electron and hole quasi-Fermi levels; E_C and E_V are the conduction and valence band edges; $J_{p,PV}$ and $J_{n,PV}$ are the hole and electron photocurrents (and are equal at steady state); $J_{p,fwd}$ is the hole current in the forward bias direction [25].

In the recent years, cQD solar cell efficiencies have increased up to 3.6% using the Schottky junction and lead chalcogenides as the QD semiconductor. However, certain

mechanisms still limit the efficiency in this type of photovoltaic cell. Indeed, for the Schottky design, both FF and V_{oc} are low for a given J_{sc} due to the poor barrier for hole injection into the electron-extracting contact. This issue can be alleviated by using the depleted heterojunction (DH) design leading to a maximized FF, V_{oc} and J_{sc} which allow this architecture to reach an efficiency of 5.1% [25].

Let us examine how the DH architectures improve the Schottky device architecture which suffers from several inherent limitations.

Firstly, the hole injection barrier at the electron-extracting electrode in the Schottky device is less effective under photovoltaic operating than the DH device, thereby promoting unwanted recombination. Let us examine how that is the case. Figure 13 shows that both the Schottky and the DH architectures exhibit a depletion region that arises from charge transfer from the electron-accepting contact to the p-type cQD film. Because metals have a high free electron density (10^{22} cm^3), there is a negligible depletion region on the metal side of the Schottky junction whereas in the DH-cQD device, the TiO_2 electrode is partially depleted since it has a much lower n-type carrier density (10^{16} cm^3) [25]. Therefore, electron backtransfer from TiO_2 to the PbS cQD in the DH architecture is strongly suppressed by the built-in potential of the depletion region. This observed built-in potential can be explained by the Fermi level difference between the n-type TiO_2 and the p-type PbS cQD film.

Secondly, in Schottky configurations, defect states at the metal–semiconductor interface can lead to Fermi-level pinning which fixes the Schottky barrier height Φ_B at the metal–semiconductor junction. As a result, the built-in potential qV_{oc} is limited to a value lower than Φ_B and thus the achievable open-circuit voltage is now constrained as well. However, the TiO_2–cQD interface in the DH structure can be effectively passivated during the solution-phase deposition of the cQDs. Therefore, it is possible to reduce interfacial trap states so that the electronic transport is better for the DH structure than the Schottky configuration.

Thirdly, most light absorption does not occur at the metal–semiconductor junction for a Schottky device but happens deeper in the device. This means that photogenerated carriers are created mainly outside the depletion region which is confined near the metal–semiconductor junction. Therefore, the carriers must traverse the entire film thickness to reach the collecting electrode. This long diffusion path significantly increases the probability of carrier recombination before reaching the electrode. This leads to a reduced photocurrent and a lower open-circuit voltage. In contrast, the DH design does not suffer from this issue because the built-in electric field inside the depletion region extends throughout the light-absorbing region. Therefore, photogenerated carriers are separated and collected by drift where they are created. This minimizes recombination losses and ensure efficient carrier separation and collection which no longer rely on diffusion.

In the following paragraph, we investigate in greater detail how QD size influences charge-carrier transport.

For this purpose, a heterojunction is constructed between a transparent TiO_2 electrode and an active layer composed of PbS cQDs of varying diameters. The energy level of these materials are shown in Figure 14. "At the smallest size (3.7 nm), the 1S electronic excited state in the PbS cQDs lies above (∼0.3 eV) the TiO_2 conduction band level and photoexcited electron injection into TiO_2 would be energetically favorable" [25]. The 1P hole excited state in the cQDs shows a large (∼2.2 eV) discontinuity with the TiO_2 valence band which induce a large barrier to the unwanted passage of majority holes

from the p-type cQD layer into the TiO_2 electrode. As the cQD diameter increases, the hole injection barrier also increases, while the driving force for electron transfer into TiO_2 progressively decreases, becoming negligible at a diameter of 5.5 nm [25]. In Figure 15, the experimental V_{oc} values are shown above each peak and the upper limit to V_{oc}, calculated from the difference in quasi-Fermi levels between the cQD layer and the TiO_2 electrode, is shown as a dashed line. This figure illustrates the increase in recombination processes because of the diminution of the built-in electric field as the QD becomes bigger.

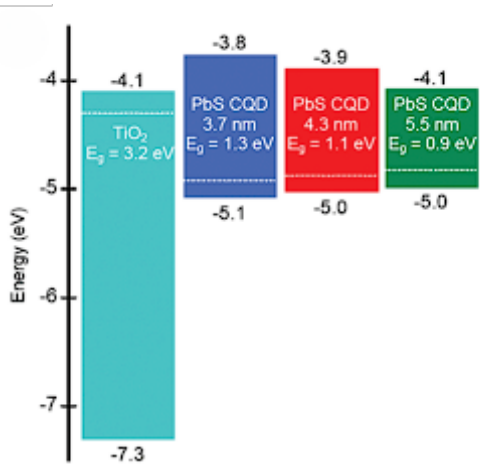


Figure 14: Energy level alignment of TiO_2 and PbS cQD of different sizes. The Fermi level is shown as a dashed line for 10^{16} cm^{-3} n-type doping in TiO_2 and 2.10^{16} cm^{-3} p-type doping in PbS cQD [25].

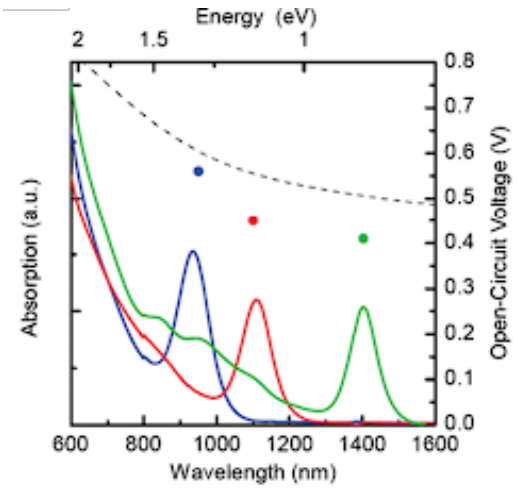


Figure 15: Solution absorption spectra in toluene of the three different PbS cQD sizes [25].

Previously, we saw that the largest cQD films of diameter 5.5 nm have a negligible conduction band offset relative to TiO_2. Yet, we notice in Figure 16 that these 0.9 eV quantum-confined bandgap devices still show a short-circuit current and a V_{oc} of 0.38 V [25]. These observations suggest that a large conduction band discontinuity between the PbS cQD and the TiO_2 semiconductor is not a requirement for efficient electron transfer. This means that the making of highly efficient devices could become easier as the requirement of a large band offset between the electron donor and acceptor can impose a substantial efficiency constraint on conventional solar cells.

To summarize, by integrating an efficient hole-blocking mechanism through a heterojunction architecture and maintaining a low minority carrier density via depletion in the wide-bandgap region, it becomes possible to simultaneously optimize all three key photovoltaic performance parameters : J_{sc}, V_{oc} and FF. This optimization is not attainable through architectures relying solely on exciton diffusion or simple Schottky metal contacts.

4.1.2 Intermediate Band Solar Cells

Among ultrahigh efficiency solar cells technology that exceeds the equivalent single p-n junction solar cell, there is a particular QD-based technology called intermediate band solar cell which captures a greater fraction of the incident solar spectrum via an engineered

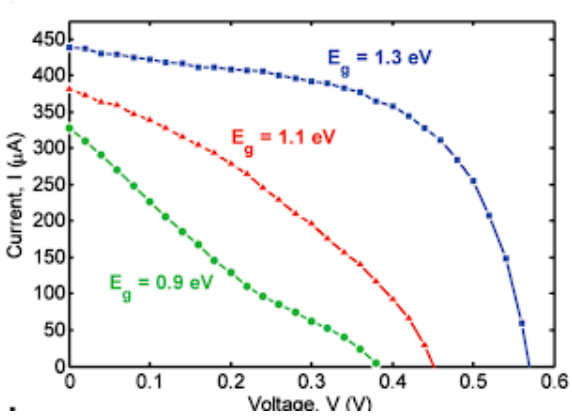


Figure 16: The photovoltaic response of the bandgap-tuned DH solar cells [25].

narrow band of electronic states within the semiconductor band gap. QD-IBSC can be done by incorporating an array of self-assembled InAs semiconductor QDs within the intrinsic region of a GaAs p-i-n diode. It enables additional sub-bandgap photon absorption pathways which also can promote an electron from the valence band to the conduction band and thereby raise power conversion efficiency. By precisely controlling the shape and size of the QDs in the quantum dot array, we have a way to engineer the discrete energy levels within the QDs and ultimately the separation between the IB and the CB.

For the IBSC to function correctly, the semiconductor VB, IB and CB must all be electronically isolated. In order to do that, we need to have well-separated quasi-Fermi levels under external illumination and zero density of states between the IB and CB.

The well-separated quasi-Fermi levels can be achieved by changing the shape of QDs. Indeed, we see in Figure 18 that the quantum engineered QD-IBSC exhibits a clear energy separation between e0 and e1 states in contrast with the reference QD-IBSC in Figure 20 [26]. This means that the quantum engineered QD-IBSC provide much better conditions for the formation of the quasi-Fermi level separation between the IB and CB under external illumination.

The zero density of states between the IB and CB can also be achieved by the engineered QD-IBSC. Indeed, we see in Figure 21a, corresponding to the quantum engineered array, clear regions of zero DOS between e1 and e0 which delimit the critical energy gap. This is the experimental signature of zero DOS between IB and CB [26]. Figure 21a also demonstrates how quantum engineering enhances absorption of lower energy photons by enabling additional transitions via the intermediate band. In contrast, there are not any sharp absorption peaks in Figure 21b corresponding to the reference QD. This is because most transitions involve the CB continuum rather than discrete QD excited states thereby producing a smooth absorption rise instead of narrow peaks. Figure 21b also shows little effective intermediate-band contribution so that absorption behaves nearly as a conventional semiconductor.

The resulting increase in J_{sc} that can be achieved using the reference QD-IBSC comes at the expense of significant reduction in V_{oc}. This decrease of V_{oc} happens because of the loss of IB features and more specifically the loss of the quasi-Fermi levels separation between IB and CB under illumination. What we just described is a key bottleneck in the development of the QD-IBSC technology. However, studies have shown that by precisely

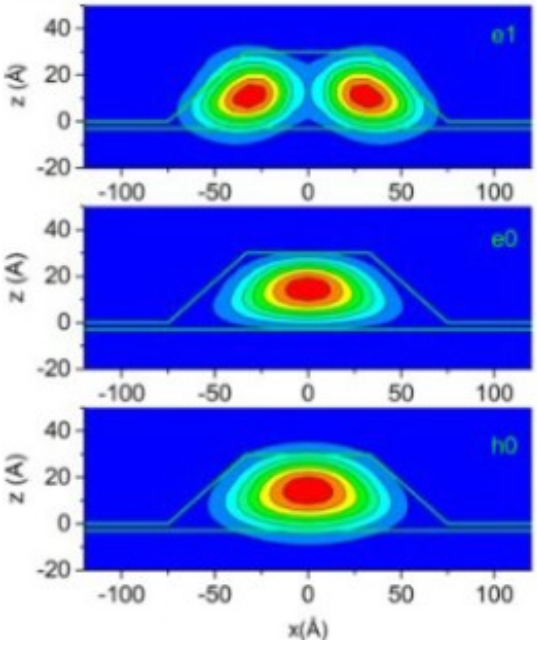


Figure 17: Charge density of a QD from a quantum engineered array (where e0 and e1 denote ground and first excited states in the CB and h0 is the ground state in the VB) [26].

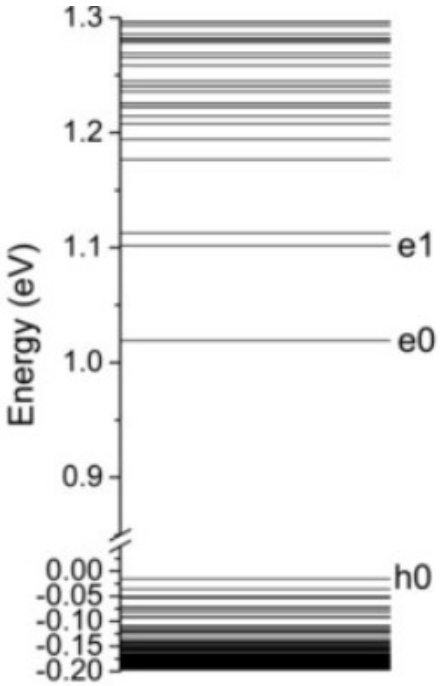


Figure 18: Electronic structure for individual engineered QDs calculated using dimensions obtained from STEM measurements [26].

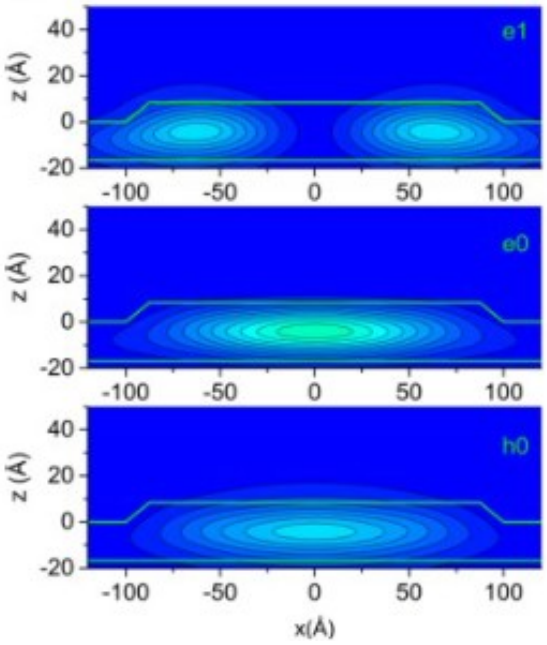


Figure 19: Charge density of a QD from a reference array (where e0 and e1 denote ground and first excited states in the CB and h0 is the ground state in the VB) [26].

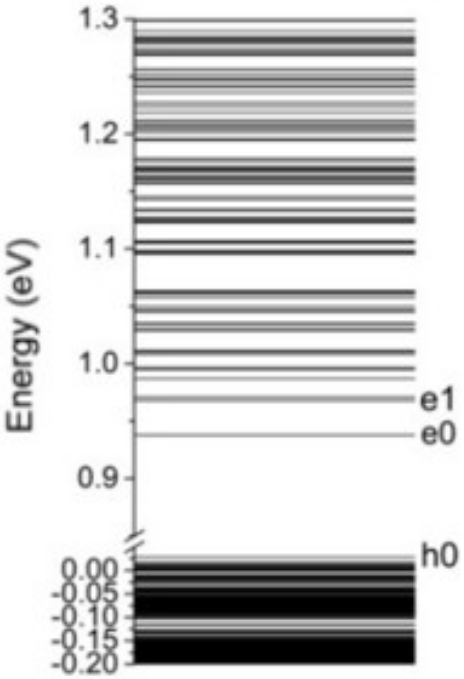


Figure 20: Electronic structure for individual reference QDs calculated using dimensions obtained from STEM measurements [26].

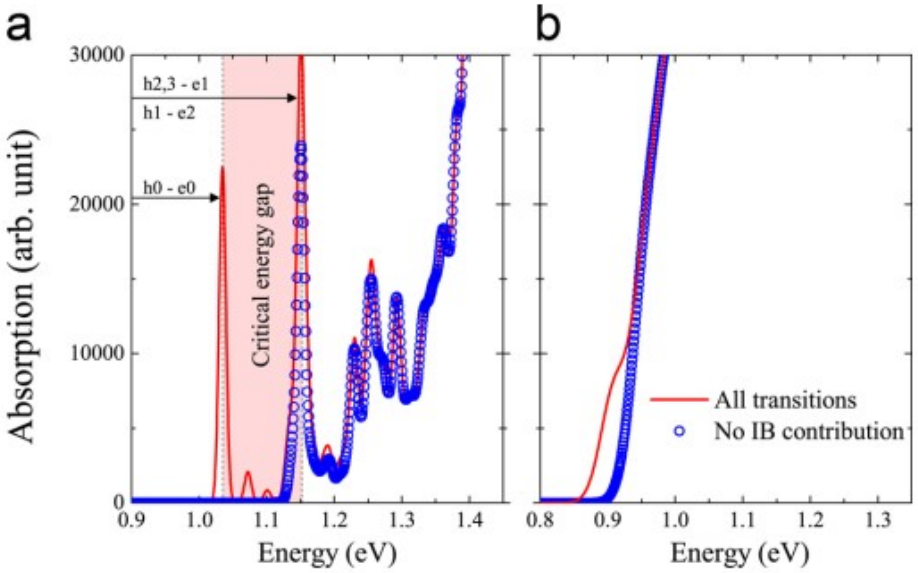


Figure 21: Absorption spectra of quantum dot arrays corresponding to (a) a quantum engineered QD-IBSC and (b) a reference QD-IBSC. The solid line accounts for optical transitions between all states in the CB and VB while the open circles indicate the case for which transitions originating from e0 have been subtracted. Labels and horizontal arrows mark the dominate transitions that contribute to particular absorption peaks [26].

engineering the shape and size of the QDs within the array it is possible to create an IB of states which is energetically isolated from the CB [26]. Indeed, simulations based on experimentally determined QD dimensions reveal the formation of a distinct second energy gap which in turn enables the device to maintain a high V_{oc}. Therefore we now know that nanometer variations in semiconducting QDs, which are used to create the IB inside the host semiconductor, have a profound effect on the macroscopic device performance.

Figure 22 shows that the engineered QD-IBSC increase V_{oc} to 840 mV under the 1 sun concentration [26]. In this specific case, a dual surface layer of ZnS/MgF_2 antireflection coating and a back surface AlGaAs reflector are added which resulted in J_{sc} =26mA/cm^2 and η = 16.8%. However, IBSC really begin to exhibit the potential for ultrahigh efficiency solar cells when the concentration of sunlight becomes important. Indeed, under concentrated illumination, not only does the open-circuit voltage increase more rapidly than in a single-bandgap device but the short-circuit current density also exhibits a significant enhancement. This increase in J_{sc} comes from the additional photocurrent generated via the intermediate band, which scales quadratically with the incident optical power. As shown in Figure 22, the device maintains excellent performance up to a concentration of 5 suns, with minimal degradation of the fill factor. The resulting power conversion efficiency reaches η =18.4% and further increases to a record value of 19.7% when accounting for the active area by excluding the contact regions. At illumination levels exceeding 5 suns, the fill factor begins to deteriorate which limits further performance gains from the intermediate band [26].

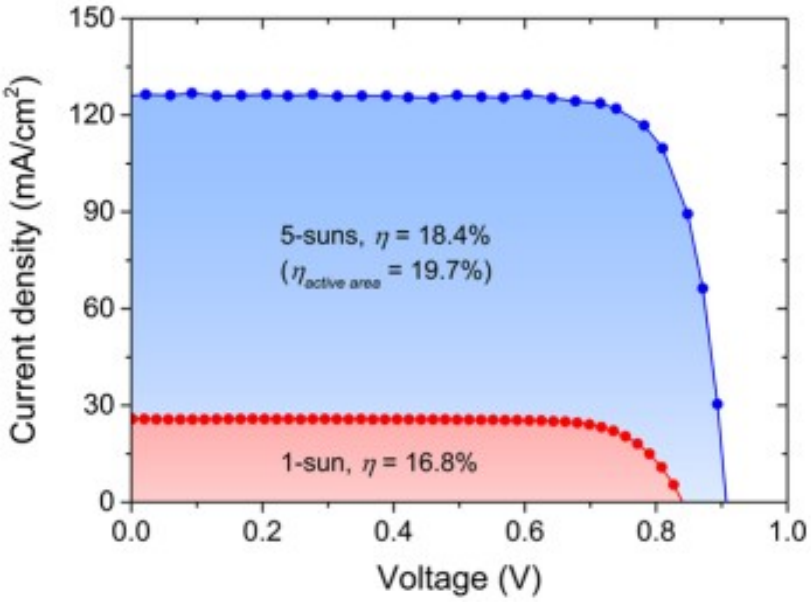


Figure 22: JV curve for an optimized quantum engineered QD-IBSC at 1 and 5 suns irradiance [26].

4.2 Single photon source

4.2.1 Principles of single photon generation from quantum Dots

The foundation of quantum dot photon generation lies in quantum confinement [13], which creates discrete, atom-like energy levels from what would otherwise be continuous energy bands in bulk semiconductors. Indeed, when a semiconductor crystal shrinks to dimensions comparable to the exciton Bohr radius, the electrons and holes experience three-dimensional spatial confinement within the finite potential well defined by the quantum dot boundaries. This confinement can be understood through the particle-in-a-box model, [16] and leads to size-tunable emission wavelengths. As stated, the bandgap energy increases as particle size decreases following the Brus equation, which accounts for both quantum confinement and Coulombic interactions. Having discussed the general form of the Brus equation in Section 2.1, we now encounter its specific expression for a spherical colloidal quantum dot [15]. It is given by

$$E_{g(\mathrm{QD})} = E_{bulk} + \frac{h^2}{8R^2}\left(\frac{1}{m_e^*} + \frac{1}{m_H^*}\right) - \frac{1.8e^2}{4\pi\varepsilon_0\varepsilon_r R^2} \tag{28}$$

with R being the radius of the quantum dot.

When a photon is absorbed by a quantum dot, its energy must match or exceed the band gap energy to excite an electron from the valence band to the conduction band. This process creates an exciton (See Figure 23), a neutral quasiparticle, formed by an electron-hole pair bound by Coulombic attraction. An exciton also has discrete energy levels, so an absorption peak similar to a δ function is present in the exciton absorption spectrum. Studying the band structure of excitons allows us to understand the different processes taking place to observe photon emission.

At zero temperature, the material band is structured with a filled valence band and an empty conduction band. Between the top of the valence band and the bottom of the

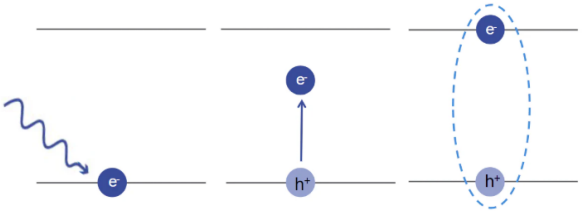


Figure 23: Formation of an exciton [27].

conduction band, there is a gap, with no electrons, called the band gap. Within the Kane model framework, the top of the valence band corresponds to an atomic-like $l = 1$, thus p state and spin-orbit coupling then yields a fourfold degenerate

$$(j = l \oplus s = 3/2,\ m_j = \pm 1/2,\ \pm 3/2) \tag{29}$$

commonly known as heavy hole,

$$\left|\frac{3}{2}, \pm\frac{3}{2}\right\rangle \tag{30}$$

and light hole,

$$\left|\frac{3}{2}, \pm\frac{1}{2}\right\rangle \tag{31}$$

Due to spatial limitations and the relevance of effective mass, the absorption of light close to the energy gap comes from a major contribution of the heavy holes. An electron, when excited from the valence band into the conduction band, will leave a similar positively charged hole. Driven by Coulomb attraction between the hole and the electron, the two are combined in an exciton state. Binding energy released in the formation of the exciton is only ≈ 5 meV in bulk material whereas the effect of confinement and spatial limitations can increase the binding energy of the excited up to 20 meV for QD structure. An explanation of the process is that there is a compression of the positive and negative electron distance which consequently strengthens the Coulomb attraction. The exciton band energy is equal to semiconductor band gap E_g, coupled with the binding energy of the electron and hole, E_e, E_h and minus the binding energy necessary to form excitons E_b :

$$E_{\text{ex}} = E_g + E_e + E_h - E_b \tag{32}$$

The binding energy of the exciton was computed in Section 2.4.

4.2.2 Absorption spectra : from bulk to quantum confinement

At room temperature, Figure 24 displays the linear absorption spectra for two CdS quantum dot samples embedded in glass. The spectrum marked "bulk" corresponds to microcrystallites in glass that are large enough to exhibit the three-dimensional properties typical of bulk CdS, featuring a broad, structureless continuum. The spectrum labeled "QD", which refers to crystals of smaller dimensions, shows clear signs of quantum confinement: in this sample, the absorption band has shifted to higher energies, exactly as expected when quantum dots experience size-dependent confinement effects.

The absorption spectra for comparable samples measured at T = 10 K are illustrated in Figure 25. At this lower temperature, phonon broadening is reduced, resulting in

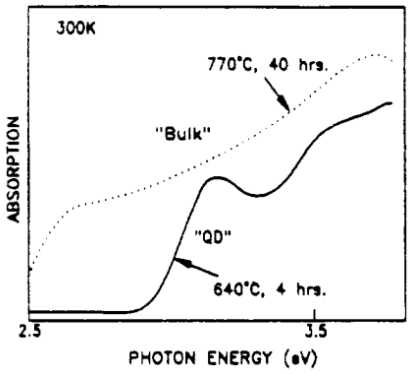


Figure 24: The linear absorption spectra at room temperature of two samples of CdS microstructures in glass [28].

narrower transitions. This allows the quantum-confined transitions to stand out more distinctly. The figure presents data for three different samples, each exhibiting a distinct quantum dot size.

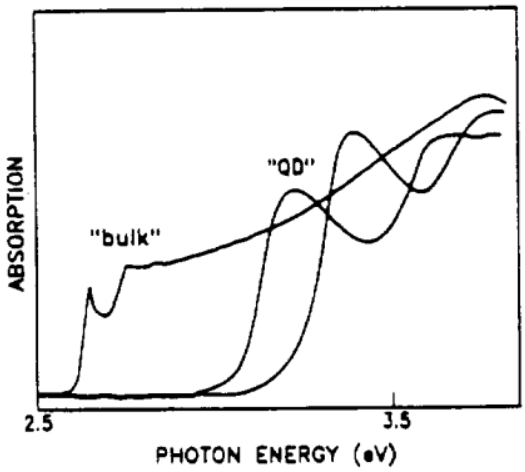


Figure 25: The linear absorption spectra at T = 10 K of CdS crystallites in glass [28].

4.2.3 Optical selection rules and light-matter interaction

When the electron and hole in the exciton recombine, a photon is emitted that is equal to the energy difference between valence and conduction states. This process is named radiative recombination and can be understood through the quantum optics concept of spontaneous emission, where a quantum system, in an excited state decays to a lower-energy state by emitting a photon, even without external driving. The spontaneous emission rate is governed by Fermi's golden rule,

$$\Gamma_{sp} = \frac{\omega^3 |d|^2}{3\pi\varepsilon_0 \hbar c^3} \tag{33}$$

where ω is the emission frequency, ε_0 is the vacuum permittivity and d is the dipole moment matrix element. By gaining control on the dipole moment, it is possible to enhance light-matter interactions and therefore lead to substantial spontaneous emission

rates. The emitted photon energy matches the band gap energy of the quantum dot, including all confinement effects.

Optical selection rules represent fundamental constraints on which electronic transitions between quantum states are allowed or forbidden. These rules arise from the symmetry properties of the quantum dot system and the electromagnetic field coupling. For a spherical (colloidal) quantum dot, the optical dipole transition matrix element for an electric dipole (E1, allowed) transition can be written as

$$d_{fi} = \langle \Psi_f | \mathbf{d} | \Psi_i \rangle \tag{34}$$

where $\mathbf{d}$ is the electric dipole moment operator. For this matrix element to be non-vanishing, it requires to be symmetric under its relevant symmetry group. Those considerations can be summed up in the so-called selection rules :

$$\Delta L = \pm 1 \qquad \Delta S = 0 \qquad \Delta J = 0, \pm 1 \qquad (J = 0 \nleftrightarrow J = 0) \tag{35}$$

Where L is the orbital angular momentum, S is the spin and $J = L+S$ is the total angular momentum. It is important to note that these rules arise from the spherical symmetry of the QD potential as well as the parity properties of the dipole operator. Specifically, transitions between quantum states of the same parity are forbidden.

4.2.4 Multi excitons effects, Auger recombination and photon statistics

The essential reason a quantum dot can act as a single photon emitter is that radiative recombination is inherently a one exciton process: a single electron-hole pair recombines to release exactly one photon. This direct correspondence underpins the ability of quantum dots to generate individual photons. But the fundamental challenge for quantum dot single photon sources arises when the quantum dot contains multiple excitons simultaneously. Suppose a biexciton (two electron-hole pairs) exists in the same quantum dot, they can potentially recombine sequentially, emitting two photons and thus degrading the single photon purity. Here is where Auger recombination [29] [4], normally a nuisance for light emission efficiency, becomes a tremendous asset for single photon generation. While radiative biexciton decay occurs on nanoseconds timescales, Auger recombination of biexcitons is extraordinarily fast; typically 10-100 picoseconds in colloidal quantum dots [4]. Auger recombination is a non-radiative recombination, where the electron-hole recombination energy is dissipated as heat without emitting a photon. The physical sequence involes

1. An electron in the conduction band recombines with a hole in the valence band
2. The recombination energy is transferred to a third charge carrier
3. This carrier is excited to a higher energy state
4. Then it gives up its additional energy in a series of collisions with the lattice, relaxing back to the edge of the band.

The efficiency of Auger recombination depends critically on the quantum dot size, smaller dots have higher Auger rates due to increased carrier overlap and stronger confinement. Hence, it is possible to "control" the Auger recombination by manufacturing accordingly the QDs.

This so-called single photon purity [30] can be expressed using the second-order photon correlation function $g^{(2)}(\tau)$. This function measures the probability of detecting a second photon at time $t+\tau$ given that a photon was detected at time t,

$$g^{(2)}(\tau) = \frac{\langle n(t)n(t+\tau)\rangle}{\langle n(t)\rangle^2} \tag{36}$$

where $n(t)$ is the photon number at time t. This quantity is basically the probability of detecting two simultaneous photons, normalized by the probability of detecting two photons at once for a random photon source.
For an ideal single photon source, we would expect $g^{(2)}(0) = 0$, meaning it is impossible to measure two photons at the same time. This phenomenon is known as (perfect) antibunching. Classical light sources such as thermal sources or coherent lasers exhibit $g^{(2)}(0) \geq 1$, indicating that photons arrive randomly or in bunches. On the other hand, quantum dots, operating as single photon emitters, exhibit strong antibunching with $g^{(2)}(0) < 0.05$ [4] [31] achievable at room temperature. The Figure 26 shows $g^{(2)}(\tau)$ depending on the interphoton delay for (I) thermal, (II) coherent (laser) and (III) antibunched (QDs) light sources.

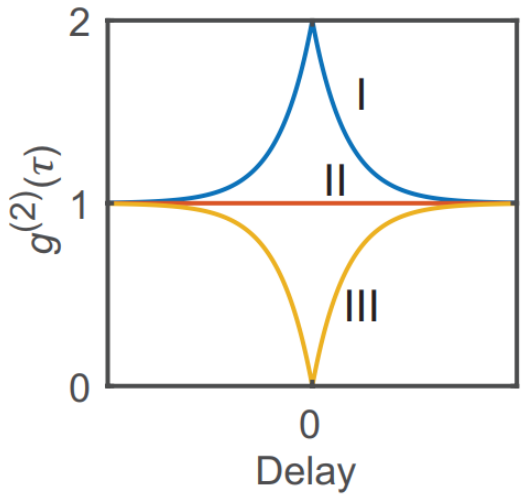


Figure 26: The second-order coherence function $g^{(2)}(\tau)$ versus interphoton delay.

Experimentally, $g^{(2)}(\tau)$ is measured using a Hanbury-Brown-Twiss setup [4]. The setup, depicted in Figure 27, consists of a light source, a beam splitter, two photodetectors, and a coincidence counter. The light from the source is split into two paths by the beam splitter and each path is directed towards a photodetector. The photodetectors are connected to a coincidence counter, which measures the correlation between the arrival times of photons at the two detectors Using this device, we can experimentally measure the second-order photon correlation.

4.2.5 A challenge for quantum dots : blinking

A critical limitation of, for example, simple CdSe quantum dots is that while Auger recombination does suppress biexciton emission, it also competes with single exciton radiative recombination. Hence, reducing the overall brightness. Additionally, the strong confinement in "small" dots creates inherent blinking [4] [32], *i.e.* random photon emission

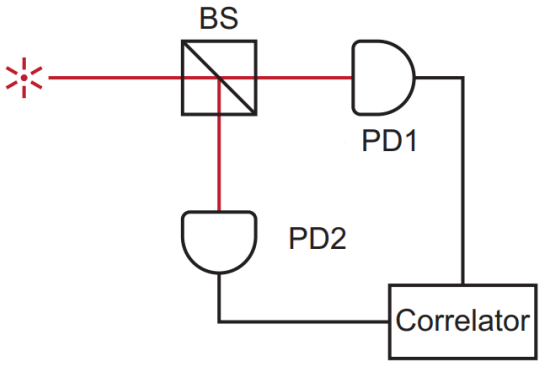


Figure 27: The Hanbury-Brown-Twiss setup for photon statistics.

interruptions [33]. This typically occurs when one of the photoexcited carriers (an electron or hole) escapes to the surface, leaving behind a charged nanocrystal. In this state, non-radiative Auger recombination dominates and quenches photoluminescence, turning the quantum dot "off". But it is possible to significantly reduce or even eliminate the blinking process in quantum dots through materials engineering strategies. By growing thick shells QDs [34] [35], it is possible to physically reduce surface trap states that cause blinking, leading to nearly non-blinking quantum dots that can spend a large amount of time in a so-called bright state.

4.2.6 Discussion on single photon generation, materials and limitations

Colloidal quantum dot as single photon sources have achieved remarkable performance metrics approaching requirements for practical quantum technologies [36] [37]. Electrically driven single photon sources [31] based on on CdSe/CdS core/shell quantum dots demonstrate near-optimal antibunching with $g^{(2)}(0) < 0.05$ at room temperature. Colloidal InP QDs have emerged as environmentally-friendly alternatives to cadmium-based materials, achieving highly stable and pure single photon emission at room temperature. Indeed, InP/ZnSe core/shell quantum dots [38] [39] exhibit strong photon antibunching with $g^{(2)}(0) < 0.09$. A landmark field trial demonstrated quantum key distribution using deterministic quantum dot single photon sources over an 18km fiber in the center of Copenhagen [40], achieving secret key rates exceeding 2 kbit/s with remarkable stability over 24 hours of continuous operation. This experiment *validated* the maturity of quantum dot technology for real-world quantum communication networks.

Before ending this section, it should be noted that quantum dots also pose a material challenge regarding its toxicity. Indeed, the presence of heavy metals, particularly cadmium (and lead), in quantum dots poses environmental and health risks that constrain widespread deployment. Cadmium has a biological half-life in the range of 15-35 years and is also classified as a probable carcinogen. Environmental regulations including Restrictions of Hazardous Substances (RoHS) in Europe and similar policies around the world restrict the growth of cadmium-based materials. Hence, there is a profound need of cadmium-free QD alternatives. In that aspect, indium phosphide and zinc-based materials have progressed substantially, catching on CdSe/CdS-based QDs performances.

5 Conclusion

In this report, we have discussed the important role of quantum confinement in understanding the unique properties of quantum dots. Their nanometer-scale dimensions lead to discrete energy levels, therefore enabling precise control over absorption and emission processes simply by tuning their size, shape, and also the core/shell structure. These features not only underline their fundamental interest in nanophysics but also explain the wide range of technological applications they enable.
Colloidal synthesis, based on mechanisms such as the LaMer model and the hot-injection method, allows the fabrication of nanocrystals with exceptional dimensional control, a key requirement for linking theoretical predictions to experimental performance. We also showed that these materials open the way to major advancements in photovoltaics, whether through cQD heterojunction devices or intermediate band solar cells, both capable of overcoming some of the limits of conventional semiconductor solar cells. Moreover, quantum dots serve as a powerful platform for single-photon generation thanks to their atom-like electronic structure and the tunability of dynamics associated to excitons.

Despite these advantages, several challenges remain, including the suppression of non-radiative recombination, the mitigation of blinking, and the transition to environmental-friendly materials. However, recent progress demonstrates that these issues can be addressed through materials engineering. Quantum dots therefore stand today as essential tools, both for deepening our understanding of quantum physics and for driving emerging technologies in energy, optoelectronics and quantum information science.

References

[1] M. A. Reed, R. T. Bate, K. Bradshaw, W. M. Duncan, W. R. Frensley, J. W. Lee, and H. D. Shih. Spatial quantization in GaAs–AlGaAs multiple quantum dots. *Journal of Vacuum Science & Technology B: Microelectronics Processing and Phenomena*, 4(1):358–360, January 1986.

[2] J. Cibert, P. M. Petroff, G. J. Dolan, S. J. Pearton, A. C. Gossard, and J. H. English. Optically detected carrier confinement to one and zero dimension in GaAs quantum well wires and boxes. *Applied Physics Letters*, 49(19):1275–1277, November 1986.

[3] K. Kash, A. Scherer, J. M. Worlock, H. G. Craighead, and M. C. Tamargo. Optical spectroscopy of ultrasmall structures etched from quantum wells. *Applied Physics Letters*, 49(16):1043–1045, October 1986.

[4] Yongzheng Ye, Xing Lin, and Wei Fang. Room-Temperature Single-Photon Sources Based on Colloidal Quantum Dots: A Review. *Materials*, 16(24):7684, December 2023.

[5] Betty Butterfly. Quantum Dots: The Basics. https://www.findlight.net/blog/quantum-dots-basics/. Online; accessed on 11/11/2025.

[6] Uwe Thomaschky. Quantum Sensing. https://www.cronologic.de/applications/quantum-sensing. Online; accessed on 14/11/2025.

[7] Jaehoon Kim, Jeongkyun Roh, Myoungjin Park, and Changhee Lee. Recent Advances and Challenges of Colloidal Quantum Dot Light-Emitting Diodes for Display Applications. *Advanced Materials*, 36(20):2212220, 2024.

[8] L. E. Brus. Electron–electron and electron-hole interactions in small semiconductor crystallites: The size dependence of the lowest excited electronic state. *The Journal of Chemical Physics*, 80(9):4403–4409, May 1984.

[9] Christopher B. Murray, David J. Norris, and Moungi G. Bawendi. Synthesis and characterization of nearly monodisperse cde (e = sulfur, selenium, tellurium) semiconductor nanocrystallites. *Journal of the American Chemical Society*, 115:8706–8715, 1993.

[10] Aldona Sashchiuk, Diana Yanover, Anna Rubin-Brusilovski, Georgy I. Maikov, Richard K. Čapek, Roman Vaxenburg, Jenya Tilchin, Gary Zaiats, and Efrat Lifshitz. Tuning of electronic properties in iv–vi colloidal nanostructures by alloy composition and architecture. *Nanoscale*, 5:7724–7745, 2013.

[11] Paul Davies. Quantum Waves — Introduction to Quantum Physics. https://www-users.york.ac.uk/ pjd113/intro.html, 2022. Online; accessed on 16/11/2025.

[12] Prof Dr A. Dayalan. 12-(QC-3): Quantum Chemistry (Particle in a box & Quantum Effects). https://dradchem.wordpress.com/2015/09/05/12-qc-3-quantum-chemistry-particle-in-a-box-quantum-effects/, September 2015.

[13] Lucjan Jacak, Arkadiusz Wójs, and Paweł Hawrylak. *Quantum Dots*. Springer, Berlin, Heidelberg, 1998.

[14] Yarub Al-Douri. *Nanomaterials: Basics to Applications*. Springer Nature Singapore, Singapore, 2022.

[15] Ephrem O. Chukwuocha, Michael C. Onyeaju, and Taylor S. T. Harry. Theoretical Studies on the Effect of Confinement on Quantum Dots Using the Brus Equation. *World Journal of Condensed Matter Physics*, 02(02):96–100, 2012.

[16] Prof. Subhasis Ghosh. Confinement of a particle in a box and in 3D (Quantum Dot) – Physics at nanoscale I. https://ebooks.inflibnet.ac.in/phy12/chapter/confinement-of-a-particle-in-a-box-and-in-3d-quantum-dot/. Online ; accessed on 11/11/2025.

[17] Emily Magee. The Five Methods for Synthesizing Colloidal Quantum Dots. https://www.azonano.com/article.aspx?ArticleID=6646. Online ; accessed on 11/11/2025.

[18] Victor K. LaMer and Robert H. Dinegar. Theory, production and mechanism of formation of monodispersed hydrosols. *Journal of the American Chemical Society*, 72(11):4847–4854, 1950.

[19] Anand Parkash, Mazharul Islam, Abdul Majeed Pirzada, and Khan Muhammad Qureshi. Bimetallic Gold-Palladium Nanoparticles: Applications in Methanol and Ethanol Oxidation Reactions. *ECS Journal of Solid State Science and Technology*, 11(10):101005, October 2022.

[20] Hiromasa Yagyu, Mao Hamamoto, and Yuanwei Wang. Analyzing the critical mixing time for the liquid-phase reduction synthesis of monodisperse gold nanoparticles using glass microfluidics. *Microfluidics and Nanofluidics*, 26(2):8, February 2022.

[21] Maryna I. Bodnarchuk and Maksym V. Kovalenko. Engineering colloidal quantum dots: Synthesis, surface chemistry, and self-assembly. In Gerasimos Konstantatos and Edward H. Sargent, editors, *Colloidal Quantum Dot Optoelectronics and Photovoltaics*, pages 1–29. Cambridge University Press, 1 edition, November 2013.

[22] Alan S. Edelstein and Robert C. Cammarata, editors. *Nanomaterials: Synthesis, Properties and Applications*. Institute of Physics Publication, 2002.

[23] A. C. Lokhande, V. C. Lokhande, D. S. Dhawale, I. A. Qattan, Shashikant Patole, and Chandrakant D. Lokhande. Optically Active Metal Oxides for Photovoltaic Applications. In Fabian I. Ezema, Chandrakant D. Lokhande, and Rajan Jose, editors, *Chemically Deposited Nanocrystalline Metal Oxide Thin Films*, pages 165–195. Springer International Publishing, Cham, 2021.

[24] Hau-Vei Han, Chien-Chung Lin, Yu-Lin Tsai, Hsin-Chu Chen, Kuo-Ju Chen, Yun-Ling Yeh, Wen-Yi Lin, Hao-Chung Kuo, and Peichen Yu. A Highly Efficient Hybrid GaAs Solar Cell Based on Colloidal-Quantum-Dot-Sensitization. *Scientific Reports*, 4(1):5734, July 2014.

[25] Andras G. Pattantyus-Abraham, Illan J. Kramer, Aaron R. Barkhouse, Xihua Wang, Gerasimos Konstantatos, Ratan Debnath, Larissa Levina, Ines Raabe, Mohammad K. Nazeeruddin, Michael Grätzel, and Edward H. Sargent. Depleted-Heterojunction Colloidal Quantum Dot Solar Cells. *ACS Nano*, 4(6):3374–3380, June 2010.

[26] Neil S. Beattie, Patrick See, Guillaume Zoppi, Palat M. Ushasree, Martial Duchamp, Ian Farrer, David A. Ritchie, and Stanko Tomić. Quantum Engineering of InAs/GaAs Quantum Dot Based Intermediate Band Solar Cells. *ACS Photonics*, 4(11):2745–2750, November 2017.

[27] Mary O'Kane. Exciton: An Introduction. https://www.ossila.com/pages/what-is-an-exciton. Online; accessed on 11/11/2025.

[28] Donglu Shi, Zizheng Guo, and Nicholas Bedford. Semiconductor Quantum Dots. In *Nanomaterials and Devices*, pages 83–104. Elsevier, 2015.

[29] Auger Recombination - an overview | ScienceDirect Topics. https://www.sciencedirect.com/topics/engineering/auger-recombination. Online ; accessed on 11/11/2025.

[30] Pascale Senellart, Glenn Solomon, and Andrew White. High-performance semiconductor quantum-dot single-photon sources. *Nature Nanotechnology*, 12(11):1026–1039, November 2017.

[31] Xing Lin, Xingliang Dai, Chaodan Pu, Yunzhou Deng, Yuan Niu, Limin Tong, Wei Fang, Yizheng Jin, and Xiaogang Peng. Electrically-driven single-photon sources based on colloidal quantum dots with near-optimal antibunching at room temperature. *Nature Communications*, 8(1):1132, October 2017.

[32] Sander J. W. Vonk and Freddy T. Rabouw. Biexciton Blinking in CdSe-Based Quantum Dots. *The Journal of Physical Chemistry Letters*, 14(23):5353–5361, June 2023.

[33] Alexander L. Efros and David J. Nesbitt. Origin and control of blinking in quantum dots. *Nature Nanotechnology*, 11(8):661–671, August 2016.

[34] Sergii Morozov, Stefano Vezzoli, Alina Myslovska, Alessio Di Giacomo, N. Asger Mortensen, Iwan Moreels, and Riccardo Sapienza. Purifying single photon emission from giant shell CdSe/CdS quantum dots at room temperature. *Nanoscale*, 15(4):1645–1651, 2023.

[35] Hao A. Nguyen, Benjamin F. Hammel, David Sharp, Jessica Kline, Griffin Schwartz, Samantha Harvey, Emily Nishiwaki, Soren F. Sandeno, David S. Ginger, Arka Majumdar, Sadegh Yazdi, Gordana Dukovic, and Brandi M. Cossairt. Colossal Core/Shell CdSe/CdS Quantum Dot Emitters. *ACS Nano*, 18(31):20726–20739, August 2024.

[36] Paweł Holewa, Aurimas Sakanas, Ugur M. Gür, Paweł Mrowiński, Alexander Huck, Bi-Ying Wang, Anna Musiał, Kresten Yvind, Niels Gregersen, Marcin Syperek, and Elizaveta Semenova. Bright Quantum Dot Single-Photon Emitters at Telecom Bands Heterogeneously Integrated on Si. *ACS Photonics*, 9(7):2273–2279, July 2022.

[37] Cherie R. Kagan, Lee C. Bassett, Christopher B. Murray, and Sarah M. Thompson. Colloidal Quantum Dots as Platforms for Quantum Information Science. *Chemical Reviews*, 121(5):3186–3233, March 2021.

[38] Andrew H. Proppe, David B. Berkinsky, Hua Zhu, Tara Šverko, Alexander E. K. Kaplan, Jonah R. Horowitz, Taehyung Kim, Heejae Chung, Shinae Jun, and Moungi G. Bawendi. Highly stable and pure single-photon emission with 250 ps optical coherence times in InP colloidal quantum dots. *Nature Nanotechnology*, 18(9):993–999, September 2023.

[39] Vigneshwaran Chandrasekaran, Mickaël D. Tessier, Dorian Dupont, Pieter Geiregat, Zeger Hens, and Edouard Brainis. Nearly Blinking-Free, High-Purity Single-Photon Emission by Colloidal InP/ZnSe Quantum Dots. *Nano Letters*, 17(10):6104–6109, October 2017.

[40] Mujtaba Zahidy, Mikkel T. Mikkelsen, Ronny Müller, Beatrice Da Lio, Martin Krehbiel, Ying Wang, Nikolai Bart, Andreas D. Wieck, Arne Ludwig, Michael Galili, Søren Forchhammer, Peter Lodahl, Leif K. Oxenløwe, Davide Bacco, and Leonardo Midolo. Quantum key distribution using deterministic single-photon sources over a field-installed fibre link. *npj Quantum Information*, 10(1):2, January 2024.